

Polymer Chemie

Viscometry

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1 Theory

1.1 Viscosity

Viscosity in polymers is influenced by 5 key factors. The first being the chain length of the Polymer, the longer the chains the more viscous the fluid gets. This is due to the Intramolecular forces. The second influence is the chemical structure of the polymer, different functional groups mean different intermolecular forces. The third one is the influence of the solvent. Because different solvents have different properties. The fourth is the influence of the concentration. This means with different concentration the viscosity changes. The last but not least influence is the temperature. Because the viscosity changes with different temperatures because the movement of the molecules changes.

1.2 From viscosity to M_η

The viscosity is defined after Hagen-Poiseuille by following equation:

$$\eta = \frac{\pi \cdot r^4 \cdot t \cdot \Delta p}{8 \cdot l \cdot V} \quad (1)$$

In which the Δp is defined as:

$$\Delta p = \rho \cdot g \cdot h \quad (2)$$

In these equations r is the capillary radius, t is the time the polymer needs to flow through the capillary, Δp is the pressure difference in the capillary, l is the length of the capillary, V is the Volume, ρ is the density of the solution and h is the height of the measured part.

After analyzing different polymers four different types of Viscosities could be determined.

$$\eta_{\text{rel}} = \frac{\eta}{\eta_0} \quad (3)$$

$$\eta_{\text{sp}} = \eta_{\text{rel}} - 1 \quad (4)$$

$$\eta_{\text{red}} = \frac{\eta_{\text{sp}}}{c} \quad (5)$$

$$\eta = \frac{\ln \eta_{\text{rel}}}{c} \quad (6)$$

If equation 1 is taken into account η_{sp} is changed to:

$$\eta_{\text{sp}} = \frac{t\rho - t_0\rho_0}{t_0\rho_0} \quad (7)$$

And because of the low concentration the density of the solution is almost equal to the density of the solvent. Therefore the equation can be simplified.

$$\eta_{\text{sp}} \approx \frac{t - t_0}{t_0} \quad (8)$$

To get the viscosimideled molecular Volume the Staudinger-index is used, and it is connected via the Marc-Howink-Equation.

$$[\eta] = K \cdot \bar{M}_\eta^a \quad (9)$$

This equation is used in combination with the equatoin for the viscositynumber.

$$[\eta] = \lim_{c \rightarrow 0} (\eta_{\text{red}}) = \lim_{c \rightarrow 0} \left(\frac{\eta_{\text{sp}}}{c} \right) \quad (10)$$

So by extrapolating the results η_{red} can be determined and via this $[\eta]$ and from that $\bar{M}_\eta$ can be determined. This is also used in our case in the experiment.

2 Experimental

The three different polymer solutions were already prepared and ready to use. One pure Toluene solution and thre polymer solutions with different concentrations. all of these four solutions were analyzed with a Viscosimeter.

3 Results

4 Questions

4.1 Question 1

4.2 Question 2

4.3 Question 3

4.4 Question 4